

# The Prime-Embedded Quantum Bargmann–Segal Algorithm (PEQBSA)

## Corrected “prime-on-coefficients” formulation (with references)

The Prime-Embedded Quantum Bargmann–Segal Algorithm (PEQBSA) introduces **prime-structured modulation** into the Bargmann–Segal (Segal–Bargmann / Fock-space) phase-space formalism **by acting on the Fock/Taylor coefficients** of the holomorphic representation [1,2].

**Key correction:** in a Bargmann–Segal holomorphic state function  $\psi(z)$ , primes cannot multiply  $\psi$  “directly” via an undefined index  $n$ . Instead, prime structure must act on the **Fock/Taylor coefficients**  $\{\psi_n\}$ , i.e., on the natural discrete label  $n$  of the number (Fock) basis.

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## Structure of Prime-Embedded Quantum Bargmann–Segal Algorithm (PEQBSA)

The structure of PEQBSA includes the following components:

- Prime-Encoded Bargmann–Segal Representation
  - Prime-Modulated Coherent States and Quantum Amplitudes
  - Prime-Weighted Time Evolution and Propagation
  - Prime-Controlled Quantum State Overlap and Inner Product
  - Applications in Quantum Optics, Quantum Computing, and Phase Space Methods
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## 1. Prime-Encoded Bargmann–Segal Representation

In the Bargmann–Segal formalism, quantum states are represented by **holomorphic functions** on complex phase space [1,2]. The discrete Fock index  $n$  appears naturally through the Taylor/Fock expansion.

### Bargmann–Segal State Representation

Let  $|\psi\rangle$  be a (single-mode) bosonic quantum state with Fock expansion

$$|\psi\rangle = \sum_{n=0}^{\infty} \psi_n |n\rangle, \quad \sum_{n=0}^{\infty} |\psi_n|^2 = 1.$$

The corresponding Bargmann–Segal holomorphic function is

$$\psi(z) = \sum_{n=0}^{\infty} \psi_n \frac{z^n}{\sqrt{n!}}, \quad z \in \mathbb{C}.$$

Equivalently, the coefficients are recovered by

$$\psi_n = \frac{1}{\sqrt{n!}} \left. \frac{d^n}{dz^n} \psi(z) \right|_{z=0}.$$

## Prime-Encoded Bargmann–Segal State (prime-on-coefficients)

Define a **prime-derived weight sequence**  $w_n$  (built from primes in any desired way). The prime embedding is implemented by the **diagonal (number-basis) modulation operator**

$$M_w := \sum_{n=0}^{\infty} w_n |n\rangle\langle n|.$$

The prime-embedded state is

$$|\psi_w\rangle := M_w |\psi\rangle = \sum_{n=0}^{\infty} (w_n \psi_n) |n\rangle.$$

In Bargmann–Segal form, this is

$$\psi_w(z) := (M_w \psi)(z) = \sum_{n=0}^{\infty} w_n \psi_n \frac{z^n}{\sqrt{n!}}.$$

### Notes on physical realizability

- If  $|w_n| = 1$  for all  $n$ , then  $M_w$  is **unitary** (a physically valid closed-system gate).
- If  $|w_n| \neq 1$  (e.g.  $w_n$  grows with  $n$ ), then  $M_w$  is **non-unitary** and must be interpreted as a **filter / postselected / dissipative** operation, typically requiring truncation and renormalization.

## 2. Prime-Modulated Coherent States and Quantum Amplitudes

Coherent states play a central role in quantum optics and continuous-variable quantum information [3]. In PEQBSA, primes modulate coherent states by acting on their **Fock coefficients**.

### Coherent States

A coherent state  $|\alpha\rangle$  ( $\alpha \in \mathbb{C}$ ) has the Fock expansion [3]

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle.$$

Its Bargmann–Segal function is

$$\psi_{\alpha}(z) = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{(\alpha z)^n}{\sqrt{n!}} = e^{-\frac{|\alpha|^2}{2}} e^{\alpha z}.$$

## Prime-Embedded Coherent States

Applying the prime weight operator  $M_w$  yields the prime-modulated coherent state

$$|\alpha; w\rangle := M_w |\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} w_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle.$$

Its Bargmann–Segal function is

$$\psi_{\alpha; w}(z) = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} w_n \frac{(\alpha z)^n}{n!}.$$

## Fully worked example: Prime-indicator SNAP phase gate on a coherent state

A SNAP gate is a number-selective phase operation of the form [4,5]

$$U_{\{\phi_n\}} := \sum_{n=0}^{\infty} e^{i\phi_n} |n\rangle\langle n|.$$

### Prime-indicator phase pattern

Define the prime-indicator function

$$\chi_{\mathbb{P}}(n) := \begin{cases} 1, & n \text{ is prime} \\ 0, & n \text{ is not prime} \end{cases}$$

and choose a single phase angle  $\theta \in \mathbb{R}$  with

$$\phi_n = \theta \chi_{\mathbb{P}}(n).$$

Then the gate is

$$U_{\{\text{prime}\}}(\theta) := \sum_{n=0}^{\infty} e^{i\theta \chi_{\mathbb{P}}(n)} |n\rangle\langle n|.$$

This is exactly a prime-on-coefficients modulation with weights  $w_n = e^{i\theta \chi_{\mathbb{P}}(n)}$ .

Applying  $U_{\text{prime}}(\theta)$  gives

$$|\alpha; \text{prime}, \theta\rangle := U_{\{\text{prime}\}}(\theta) |\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} e^{i\theta \chi_{\mathbb{P}}(n)} \frac{\alpha^n}{\sqrt{n!}} |n\rangle.$$

The corresponding Bargmann–Segal function is

$$\psi_{|\alpha\rangle, \text{prime}, \theta}(z) = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{e^{i\theta \chi_{\mathbb{P}}(n)} (\alpha z)^n}{n!}.$$

A useful decomposition separates prime and non-prime indices:

$$\psi_{|\alpha\rangle, \text{prime}, \theta}(z) = e^{-\frac{|\alpha|^2}{2}} \left[ \sum_{n \notin \mathbb{P}} \frac{(\alpha z)^n}{n!} + e^{i\theta} \sum_{n \in \mathbb{P}} \frac{(\alpha z)^n}{n!} \right].$$

Equivalently,

$$\psi_{|\alpha\rangle, \text{prime}, \theta}(z) = e^{-\frac{|\alpha|^2}{2}} \left[ e^{\alpha z} + (e^{i\theta} - 1) \sum_{n \in \mathbb{P}} \frac{(\alpha z)^n}{n!} \right].$$

## Overlap (fidelity amplitude) with the original coherent state

The overlap of the **original** coherent state with the **prime-gated** coherent state is

$$\langle \alpha | \psi_{|\alpha\rangle, \text{prime}, \theta} \rangle = \langle \alpha | U_{\text{prime}}(\theta) | \alpha \rangle = \sum_{n=0}^{\infty} |c_n|^2 e^{i\theta \chi_{\mathbb{P}}(n)},$$

where  $|c_n|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}$  is a Poisson distribution with mean  $\lambda = |\alpha|^2$ . Hence

$$\langle \alpha | \psi_{|\alpha\rangle, \text{prime}, \theta} \rangle = e^{-\lambda} \sum_{n=0}^{\infty} \frac{e^{i\theta \chi_{\mathbb{P}}(n)} \lambda^n}{n!} = 1 + (e^{i\theta} - 1) P_{\text{prime}}(\lambda),$$

with

$$P_{\text{prime}}(\lambda) := e^{-\lambda} \sum_{n \in \mathbb{P}} \frac{\lambda^n}{n!} \quad \text{Poisson probability mass on prime } n.$$

The corresponding **fidelity** is  $|\langle \alpha | \psi_{|\alpha\rangle, \text{prime}, \theta} \rangle|^2$ .

More generally, the coherent-state overlap (useful for Husimi-Q computations) is

$$\langle \beta | \psi_{|\alpha\rangle, \text{prime}, \theta} \rangle = e^{-\frac{|\alpha|^2 + |\beta|^2}{2}} \sum_{n=0}^{\infty} \frac{e^{i\theta \chi_{\mathbb{P}}(n)} (\beta^* \alpha)^n}{n!}.$$

## Expected nonclassicality signature

- **Photon-number distribution is unchanged.** Because  $U_{\text{prime}}(\theta)$  is diagonal in  $|n\rangle$ , it preserves  $|c_n|^2$ . So the output still has a Poisson photon-number distribution.
- **But phase-space structure becomes non-Gaussian.** The gate introduces a number-dependent phase pattern  $e^{i\theta \chi_{\mathbb{P}}(n)}$ , which reshapes interference between Fock components.
- **Wigner interference fringes and negativity.** A coherent state has a strictly positive Gaussian Wigner function. After the prime-indicator SNAP gate, the Wigner function typically develops

**oscillatory fringes** and, for sufficiently large  $|\alpha|$  and nontrivial  $\theta$  (e.g.,  $\theta = \pi$ , a prime-index sign flip), exhibits **regions of negative Wigner density**.

A compact diagnostic expression is the displaced-parity formula [6–8]

$$W(\beta) = \frac{2}{\pi} \langle \alpha; \text{prime}, \theta | D(\beta) \Pi D(\beta)^\dagger | \alpha; \text{prime}, \theta \rangle,$$

where  $D(\beta)$  is the displacement operator and  $\Pi = (-1)^{\hat{n}}$  is the parity operator. Negativity of  $W(\beta)$  at some  $\beta$  is a direct nonclassicality signature.

### 3. Prime-Weighted Time Evolution and Propagation

Time evolution can be prime-modulated in a way that remains consistent with the prime-on-coefficients principle.

#### Quantum Time Evolution

The Schrödinger equation is

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle.$$

In the number basis  $|n\rangle$ , coefficients  $\psi_n(t)$  obey

$$i\hbar \frac{d}{dt} \psi_n(t) = \sum_{m=0}^{\infty} H_{nm}(t) \psi_m(t), \quad H_{nm}(t) = \langle n | H(t) | m \rangle.$$

#### Prime-Embedded Time Evolution (coefficient-level control)

A direct prime-structured control term is a diagonal “phase-drive” Hamiltonian

$$H_{\text{prime}}(t) := \hbar \sum_{n=0}^{\infty} \dot{\phi}_n(t) |n\rangle \langle n|,$$

where  $\phi_n(t)$  is any prime-derived phase schedule (e.g.,  $\phi_n(t) = \theta(t) \chi_{\mathbb{P}}(n)$ ).

Then the evolution operator generated by  $H_{\text{prime}}(t)$  alone is precisely the SNAP form [4,5]

$$U(t) = \sum_{n=0}^{\infty} e^{i\phi_n(t)} |n\rangle \langle n|.$$

### Gate-based (stroboscopic) propagation

PEQBSA can also be implemented as alternating free evolution and prime-phase pulses:

$$|\psi_{k+1}\rangle = U_{\text{free}}(\Delta t) U_{\text{prime}}^{(k)} |\psi_k\rangle,$$

where each  $U_{\text{prime}}^{(k)}$  is a chosen diagonal prime-derived modulation.

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## 4. Prime-Controlled Quantum State Overlap and Inner Product

Prime control should not be introduced by inserting prime weights into an integral over  $z$  without defining how the discrete index enters the holomorphic representation. Instead, prime control is expressed by applying modulation operators to the states and then using the **standard** inner product [1,2].

### Quantum State Overlap

In Bargmann–Segal form, the (standard) inner product is [1,2]

$$\langle \psi_1 | \psi_2 \rangle = \int_{\mathbb{C}} \psi_1(z)^* \psi_2(z) e^{-|z|^2} \frac{d^2 z}{\pi}.$$

In Fock coefficients, it is

$$\langle \psi_1 | \psi_2 \rangle = \sum_{n=0}^{\infty} \psi_{1,n}^* \psi_{2,n}.$$

### Prime-Embedded Quantum State Overlap

Let  $|\psi_{1,w}\rangle = M_w |\psi_1\rangle$  and  $|\psi_{2,v}\rangle = M_v |\psi_2\rangle$ . Then the prime-controlled overlap is

$$\langle \psi_{1,w} | \psi_{2,v} \rangle = \langle \psi_1 | M_w^\dagger M_v | \psi_2 \rangle = \sum_{n=0}^{\infty} \psi_{1,n}^* (w_n^* v_n) \psi_{2,n}.$$

If  $w_n = e^{i\phi_n}$  and  $v_n = e^{i\varphi_n}$ , this simplifies to

$$\langle \psi_{1,w} | \psi_{2,v} \rangle = \sum_{n=0}^{\infty} \psi_{1,n}^* e^{i(\varphi_n - \phi_n)} \psi_{2,n}.$$

# 5. Applications in Quantum Optics, Quantum Computing, and Phase Space Methods

## Quantum Optics

Prime-structured diagonal phase modulation provides a systematic way to engineer **non-Gaussian states** from classical-like inputs (e.g., coherent states) [3–5]. This enables controlled creation of phase-space interference patterns and nonclassical signatures [6–8].

## Quantum Computing

In continuous-variable and bosonic encodings, number-selective phase gates are powerful primitives [4,5]. Prime-derived phase schedules provide reproducible, structured modulation patterns for benchmarking, control design, and state engineering.

## Phase Space Representations

Because PEQBSA is formulated directly in the Bargmann–Segal / phase-space language, it connects naturally to Husimi-Q and Wigner representations [1,2]. Prime modulation yields tunable distortions of phase-space distributions and interference structure [6–8].

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# Complete Prime-Embedded Quantum Bargmann–Segal Algorithm (PEQBSA)

Here’s the complete structure of the Prime-Embedded Quantum Bargmann–Segal Algorithm (PEQBSA) in prime-on-coefficients form:

### Step 1: Prime-Encoded Bargmann–Segal States

- Choose prime-derived weights  $w_n$  (preferably  $|w_n| = 1$  for a unitary embedding).
- Define  $M_w = \sum_n w_n |n\rangle\langle n|$  and  $\psi_w(z) = \sum_n w_n \psi_n z^n / \sqrt{n!}$  [1,2].

### Step 2: Prime-Modulated Coherent States

- Start from  $|\alpha\rangle$  and apply  $M_w$ :  $|\alpha; w\rangle = e^{-|\alpha|^2/2} \sum_n w_n \alpha^n / \sqrt{n!} |n\rangle$  [3].

### Step 3: Prime-Weighted Time Evolution

- Implement a prime-derived diagonal control Hamiltonian (continuous) or SNAP-style pulses (stroboscopic) [4,5]:  $U(t) = \sum_n e^{i\phi_n(t)} |n\rangle\langle n|$ .

#### Step 4: Prime-Controlled Quantum State Overlap

- Compute overlaps via the standard inner product after modulation [1,2]:  $\langle \psi_{1,w} | \psi_{2,v} \rangle = \langle \psi_1 | M_w^\dagger M_v | \psi_2 \rangle$ .
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## 6. Advantages of PEQBSA

- **Mathematically well-defined:** primes act on the discrete Fock/Taylor index  $n$ , avoiding undefined mixing of discrete and continuous variables [1,2].
  - **Physically implementable (unitary mode):** choosing  $|w_n| = 1$  gives a bona fide quantum gate (diagonal phase control) [4,5].
  - **Controlled nonclassicality:** prime-derived number-dependent phases generate non-Gaussian phase-space interference and potential Wigner negativity [6–8].
  - **Flexible information processing:** prime-structured overlaps provide a tunable similarity/contrast mechanism between states.
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## Conclusion

The Prime-Embedded Quantum Bargmann–Segal Algorithm (PEQBSA) can be made precise by implementing prime structure as a **coefficient-level (Fock-basis) modulation**. In this corrected form, PEQBSA becomes a clear operator-theoretic procedure that translates directly into Bargmann–Segal holomorphic functions and supports concrete, testable protocols such as a prime-indicator SNAP phase gate on coherent states.

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